[date of submission]

The Editors-in-Chief
Advances in Structural Engineering
Sage

Dear Editors,

Submission of an original research article: “Observability and sensor layout for influence-area load monitoring of continuous girder bridges”

We submit the above manuscript for consideration as an Original Research Article in *Advances in Structural Engineering*. It falls within the journal’s focus on structural health monitoring for performance assessment of bridges.

Monitoring the loads that a bridge actually carries is often done with few channels, by integrating one response record and dividing by a calibration constant. On continuous girders this constant can vanish, because influence lines change sign over the piers. The manuscript gives structural engineers criteria for choosing, before installation, channels and positions that observe the load, and for judging whether their constant can be trusted when the pier stiffness is uncertain. The underlying identity and the generalised least-squares estimator are classical, and the manuscript says so; the contributions are the criteria built on them:

1. a closed-form catalogue of blind loci for five channel types on two- and three-span and haunched girders, with indicators that need no reference rigidity, showing that strain gauges are blind at every zero of the uniform-load moment and nearly blind in a band that can contain gauges set back from the piers;

2. a placement rule, optimal sensors at the extremes of the uniform-load response, together with a field finding from public moving-truck records that weighting gauges by their vehicle-free noise covariance raises the out-of-sample error above that of a plain gauge average;

3. an identifiability analysis under uncertain pier stiffness, in which the constant follows a linear-fractional (Möbius) law, applied to an in-service 55 + 90 + 55 m box girder: its side-span deflection constant is not identifiable from static acceptance data, the main-span constant is, and a pier tiltmeter is the most informative additional measurement.

A laboratory test is consistent with the identity. The field records are used to check the assumptions behind the criteria; they cannot test weight discrimination, because their reference weights vary little, and the manuscript states this. The moving-load test of the in-service bridge is designed but not yet performed, and its predicted outcomes are given so that they can be checked.

The manuscript builds on bridge weigh-in-motion work published in this journal, including the review by Yu, Cai and Deng (2016). It is about 11,000 word-equivalents with 7 figures and 4 tables; derivations, full field tables and the error budget are in the Supplementary Material.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 licence). The laboratory data, the processed data of the in-service bridge, the derived data behind every figure and table, and all analysis scripts with their console output are provided as Supplementary Material.

The manuscript is original, has not been published, and is not under consideration by any other journal. Two other papers by the authors are under review at *Structures*; they address different problems and share no data, test rig or text with this manuscript. All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Yours sincerely,

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